Entropy

I/ Introduction

1) Definition and properties

The entropy measures the amount of randomness in a distribution.

Let X be a discrete random variable. Its entropy is

$$\boxed{H(X) = \sum_{x \in \Omega} -\mathbb{P}(X=x) \log_2 \left(\mathbb{P}(X=x)\right).}$$

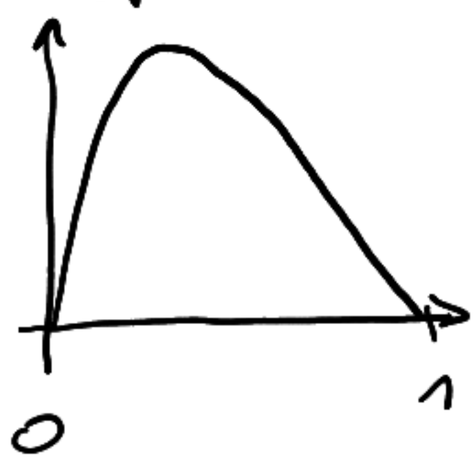
where Ω is the support of X i.e. $\{x : \mathbb{P}(X=x) > 0\}$.

- $H(X)$ measures the amount of surprises in the randomness of X.
- Alternatively, $H(X)$ measures the amount of information learned when seeing X. (Shannon source coding theorem).

Lemma 1 $H(X) \leq |\log_2(\Omega)|$ with equality iff X is uniform.

<u>Proof</u>: Let $f(p) = -p \log_2 p$

f is concave for $p \in [0,1]$.



$$H(X) = \sum_{x \in \Omega} f\left(\mathbb{P}(X=x)\right)$$

$$\underset{\text{concavity}}{\leq} |\Omega| \, f\left(\frac{1}{|\Omega|} \sum_{x \in \Omega} \mathbb{P}(X=x)\right)$$

$$\leq |\Omega| \, f\left(\frac{1}{|\Omega|}\right) = \log_2 |\Omega|. \quad \blacksquare$$

Let X, Y be random variables. Let $Z = (X, Y)$. The joint entropy of X and Y is

$$H(X,Y) := H(Z) = \sum_{x,y} -\mathbb{P}(X=x, Y=y) \log_2 (\mathbb{P}(X=x, Y=y)).$$

We write similarly $H(X_1, \dots X_n)$.

Lemma 2 (Independence). Let X and Y be independent random variables. then $H(X,Y) = H(X) + H(Y)$.

<u>Proof</u>:
$$H(X,Y) = \sum_{xy} -\mathbb{P}(X=x)\mathbb{P}(Y=y)\left[\log_2(\mathbb{P}(X=x)) + \log_2(\mathbb{P}(Y=y))\right]$$
$$= \underbrace{\sum_y \mathbb{P}(Y=y)}_{1} \cdot \sum_x -\mathbb{P}(X=x)\log_2(\mathbb{P}(X=x))$$
$$+ \sum_x \mathbb{P}(X=x) \cdot \sum_y -\mathbb{P}(Y=y)\log_2(\mathbb{P}(Y=y))$$
$$= H(X) + H(Y) \quad \blacksquare$$

Let X and Y be two random variables.

$$H(X|Y) := \mathbb{E}_y[H(X|Y=y)]$$
$$= \sum_y \mathbb{P}(Y=y)\, H(X|Y=y)$$
$$= \sum_y \mathbb{P}(Y=y) \sum_x -\mathbb{P}(X=x)\log_2(\mathbb{P}(X=x)).$$

$H(X|Y)$ measures the amount of information of X that is not already contained in Y.

Lemma 3 (Chain rule) $H(X,Y) = H(X) + H(Y|X)$.

<u>Proof</u>: Exercise session.

Observation : • If X and Y are independent $H(Y|X) = H(Y)$

• If $X = f(Y)$ then $H(X|Y) = 0$

Lemma 4 (Subadditivity): $H(X,Y) \leq H(X) + H(Y)$

| **Proof** Exercise session

Mutual information $I(X,Y) = H(X) + H(Y) - H(X,Y)$
measures the amount of information common to X and Y.

Lemma 5 : $H(X|Y,Z) \leq H(X|Z)$

| **Proof** : $H(X|Y) \underset{\text{chain rule}}{=} H(X,Y) - H(Y) \underset{\text{subadditivity}}{\leq} H(X)$

| For every z, we have $H(X|Y, Z=z) \leq H(X|Z=z)$

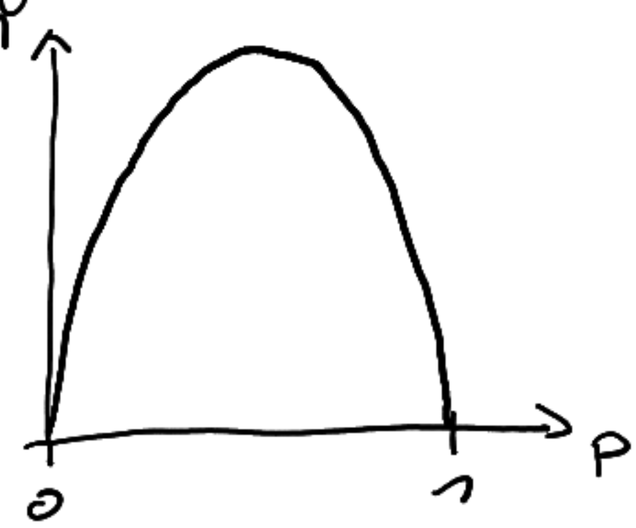
| so taking expectation on Z, we have $H(X|Y,Z) \leq H(X|Z)$ ∎

2) Usual distribution

Uniform : $X \sim \mathcal{U}(\Omega)$ $\qquad H(X) = \log_2(|\Omega|)$

Bernoulli variable : $X \sim B(p)$

$H(p) = H(X) = -p \log p - (1-p) \log(1-p)$



Binomial distribution: $X \sim Bin(n,p)$

$$H(X) = -\sum_{k=0}^{n} \binom{n}{k} p^k (1-p)^{n-k} \log_2\left[\binom{n}{k} p^k (1-p)^{n-k}\right]$$

$$\sim -\int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi\sigma}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \times \log_2\left[\frac{1}{\sqrt{2\pi\sigma}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}\right] dx$$

(where $\mu = np$ and $\sigma^2 = np(1-p)$)

$$\sim \log_2(\sqrt{2\pi\sigma}) + \frac{\sigma^2}{2\sigma^2} \log_2 e$$

$$= \frac{1}{2} \log_2(2\pi e \sigma^2)$$

de Moivre Laplace theorem (special case of the central limit theorem):

$$\binom{n}{k} p^k (1-p)^{n-k} \sim \frac{1}{\sqrt{2\pi\sigma}} e^{-\frac{(k-\mu)^2}{2\sigma^2}}$$

3) Entropy and binomial coefficients

Theorem Let $p \in [0,1]$, $n \in \mathbb{N}^+$ s.t. pn is an integer.

$$\frac{2^{nH(p)}}{n+1} \overset{(1)}{\leq} \binom{n}{np} \overset{(2)}{\leq} 2^{nH(p)}$$

<u>Proof</u>: Immediate if $p=0$ or $p=1$. $\longrightarrow$ We assume $p \in (0,1)$

- $2^{nH(p)} = 2^{-pn\log_2 p - (1-p)n\log_2(1-p)} = p^{-pn}(1-p)^{-(1-p)n}$

We have $\binom{n}{np} p^{pn}(1-p)^{(1-p)n} \leq \sum_{k=0}^{n} \binom{n}{k} p^k (1-p)^{n-k}$

$= (p + (1-p))^n = 1$ So (2).

- $\binom{n}{np} 2^{-nH(p)}$ is one of the terms of $\sum_{k=0}^{n} \binom{n}{k} p^k (1-p)^{n-k}$

we show it is the largest.

$$\binom{n}{k} p^k (1-p)^{n-k} - \binom{n}{k+1} p^{k+1} (1-p)^{n-k-1}$$

$$= \binom{n}{k} p^k (1-p)^{n-k} \underbrace{\left[1 - \frac{n-k}{k+1} \cdot \frac{p}{1-p}\right]}_{\geq 0 \text{ iff } k \geq pn-1+p}$$

As there are $n+1$ terms in $\sum_{k=0}^{n} \binom{n}{k} p^k (1-p)^{n-k}$ this proves (1) ∎

This is useful when considering sequences of biased coin tosses i.e. $X \sim Bin(n,p)$ where $p > 1/2$.

By Chernoff bound, $X \sim np$ almost always.

Hence the sequence of coin toss will almost always be one of

$\approx \binom{n}{np} \approx 2^{nH(p)}$ sequences, each occuring with probability

$\approx p^{np} (1-p)^{n(1-p)} = 2^{-nH(p)}$

Theorem: For $p \in (0, \frac{1}{2}]$ and $n \in \mathbb{N}^+$, where np is an integer,

$$\sum_{i=0}^{pn} \binom{n}{i} \leq 2^{nH(p)} = p^{pn} (1-p)^{(1-p)n}$$

<u>Proof</u>: denote $k = pn$

Let $(X_1, \dots X_n) \in \{0,1\}^n$ chosen uniformly conditioned on $\sum X_i \leq k$.

$$H(X_1, \dots X_n) \underset{\substack{\uparrow \\ \text{uniformity}}}{=} \log_2 \left(\sum_{i=0}^{k} \binom{n}{i} \right) \underset{\text{by subadditivity}}{\leq} H(X_1) + \dots H(X_n)$$

Each $X_i \sim \mathcal{B}(q)$ where $q = \mathbb{P}(X_i = 1 \mid \sum X_i = m \leq k)$

for each $m \leq k$, $\mathbb{P}(X_i = 1 \mid \sum X_i = m) = \frac{m}{n}$, so

$q \leq \frac{k}{n} = p$ Hence $H(X_i) \leq H(p)$.

Hence, $\log_2 \left(\sum_{i=0}^{k} \binom{n}{i} \right) \leq n H(p)$. ∎

This gives tails bound on binomial variables: $X \sim \mathcal{B}(n,p)$

For every $q \leq p$, $\mathbb{P}(X \leq qn) = 2^{-n} \sum_{i=0}^{\lfloor qn \rfloor} \binom{n}{i}$

so $$\frac{\log_2 \mathbb{P}(X \leq qn)}{n} \leq H(q) = -q \log \frac{q}{p} - (1-q) \log \frac{1-q}{1-p}$$

$$< 1 \quad \rightsquigarrow \quad \mathbb{P}(X \leq qn) \leq e^{-\lambda(q) n}$$

And likewise for $p \leq q \leq 1$:

$$\frac{\log_2 \mathbb{P}(X \geq qn)}{n} \leq H(q)$$

II/ Upper bounds on size of solution space.

1) Maximum number of perfect matchings.

Let G be a balanced bipartite graph on $2n$ vertices.

Denote $\phi(G)$ the number of perfect matchings of G.

We have seen in the exercise session on absorption method that for every $\varepsilon > 0$ and every large enough n,

$$\delta(G) \geq \left(\frac{1}{2} + \varepsilon\right) n \implies \phi(G) \geq \Omega(n)^{n/2}$$

In fact, the stronger result holds:

[Cuckler, Kahn 2009] : $\delta(G) \geq \frac{n}{2} \implies \phi(G) \geq \left(\frac{\delta(G)}{e + o(1)}\right)^{n/2}$

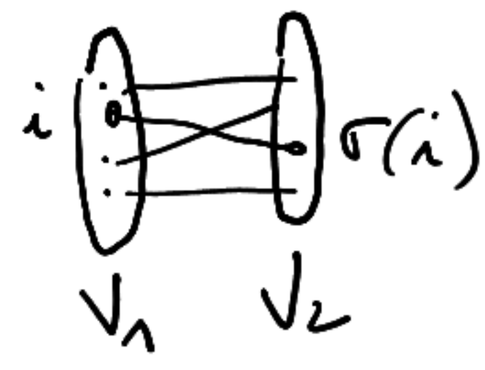
What about upper bounds on $\phi(G)$?

$\phi(G)$ is counted by the permanent of the adjacency matrix of G:

Let $V_1 \sqcup V_2$ be the bipartition of G. Let $A = (a_{ij})$ s.t.

$$a_{ij} = \begin{cases} 1 & \text{if vertex } i \text{ in } V_1 \text{ is adjacent to vertex } j \in V_2 \\ 0 & \text{otherwise.} \end{cases}$$

$$\operatorname{perm}(A) = \sum_{\sigma \in \mathfrak{S}_n} \prod_{i=1}^{n} a_{i,\sigma(i)}$$



This formula is close to the determinant: $\det(A) = \sum_{\sigma \in \mathfrak{S}_n} (-1)^{\varepsilon(\sigma)} \prod_{i=1}^{n} a_{i,\sigma(i)}$

but without the signature term.

[Conjecture by Minc 1963, Proof by Brégman 1973:]

Let $A = (a_{ij}) \in \{0,1\}^{n \times n}$. Then $\operatorname{perm}(A) \leq \prod_{i=1}^{n} (d_i!)^{1/d_i}$

where $d_i = \sum_{j=1}^{n} a_{ij}$.

<u>Observation</u>: • equality is attained if A is of the following form:

i.e. $G = K_{d_1,d_1} \sqcup \dots \sqcup K_{d_t d_t}$

• If G is δ-regular, where $\delta \sim c n$, then

$$\phi(G) \leq \prod_{i=1}^{n} (\delta!)^{1/\delta}$$
$$\leq (\delta!)^{n/\delta}$$
$$\log(\phi(G)) \lesssim \frac{n}{\delta}\left[\log(\sqrt{2\pi\delta}) + \delta \log\left(\frac{\delta}{e}\right)\right]$$
$$\lesssim n \log\left(\frac{\delta}{e}\right) + \frac{n}{\delta}\log(\sqrt{2\pi\delta})$$
$$= O\left(n \log\left(\frac{\delta}{e}\right)\right)$$

• A similar bound exists if G is not bipartite:

[Kahn Lovász] $\phi(G) \leq \prod_{v \in V(G)} (\deg(v))^{1/(2\deg(v))} = \prod_{v \in V(G)} \phi(K_{\deg v, \deg v})^{1/2\deg(v)}$

<u>Proof</u>: $\phi(G) = \sqrt{\phi(G \sqcup G)} \leq \sqrt{\phi(G \times K_2)}$ ∎

Proof of Brégman (by Radhakrishnan 1997):

Let σ be a random uniform permutation, conditioned on $a_{i\sigma(i)} = 1$ for $i \in [n]$.

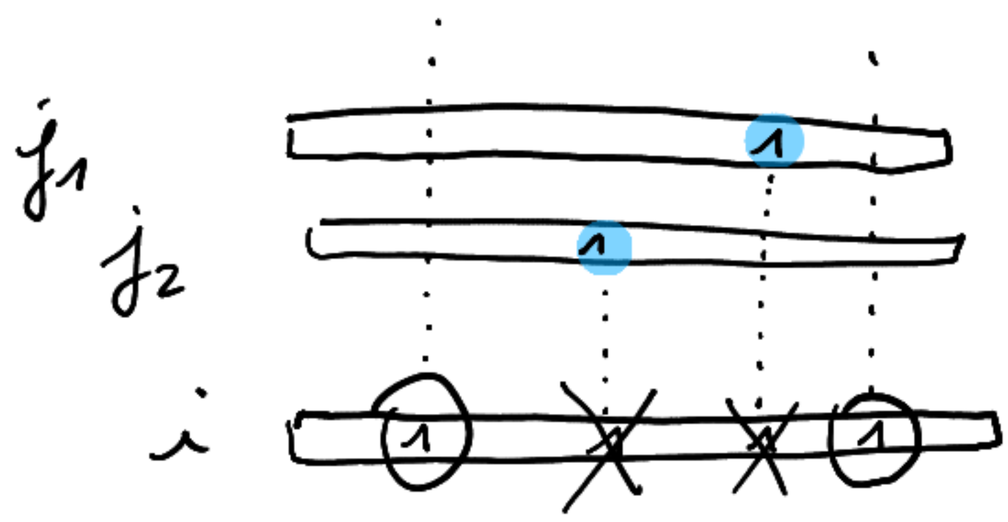
$$\begin{aligned} H(\sigma) &= \log_2(\mathrm{per}(A)) \\ &= H(\sigma(1), \sigma(2), \dots \sigma(n)) \\ &= H(\sigma(1)) + H(\sigma(2) \mid \sigma(1)) + \dots H(\sigma(n) \mid \sigma(1), \dots \sigma(n-1)) \end{aligned}$$

We have $H(\sigma(i) \mid \sigma(1), \dots \sigma(i-1)) \leq H(\sigma(i)) \leq \log_2 |\text{support } \sigma(i)|$
$\leq \log_2(d_i)$

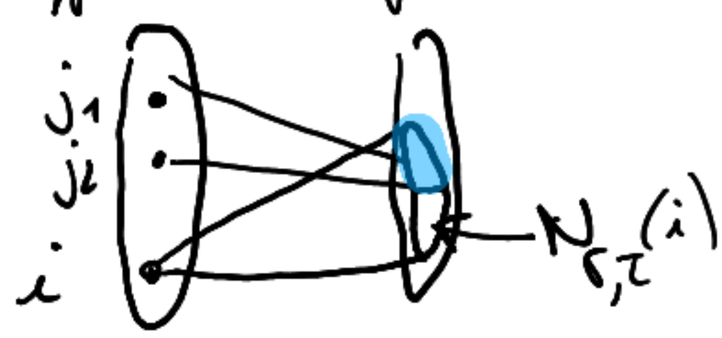
This is not good enough. Key idea: consider the $\sigma(i)$ in a random order.

$\rightsquigarrow$ Let τ be a random permutation of the columns of A.
We say j appears before i if $\tau(j) < \tau(i)$.

- Let $N_{\sigma,\tau}(i) = \#\left\{ k : a_{ik} = 1 \text{ and } \forall j \text{ appearing before } i,\ \sigma(j) \neq k \right\}$



j_1, j_2 appear before i



in other words, $N_{\sigma,\tau}(i)$ is the number of available choices to match i knowing $\{\sigma(j) : j \text{ appears before } i\}$.

$\rightarrow$ For every fixed σ, $1 \leq N_{\sigma,\tau}(i) \leq d_i$ and

Let S be the set of indices s.t. $s \in S \Leftrightarrow a_{is} = 1$

$N_{\sigma,\tau}(i)$ depends on how the rows $\sigma^{-1}(S)$ are ordered compared to i by τ. If there are exactly l rows of $\sigma^{-1}(S)$ before i, then $|N_{\sigma,\tau}(i)| = d_i - l$. Thus

$$\mathbb{P}_\tau\left(|N_{\sigma,\tau}(i)| = k\right) = \frac{1}{d_i} \qquad \forall k \in [d_i]$$

- For any fixed τ, $H(\sigma) = \sum_{i=1}^{m} H(\sigma(i) \mid \{\sigma(j) : j \text{ appears before } i\})$

chain rule

$$\leq \sum_{i=1}^{m} \mathbb{E}_\sigma \log_2 |N_{\sigma,\tau}(i)|.$$

uniform bound

If τ is a uniform permutation,

$$H(\sigma) \leq \mathbb{E}_\tau\left[\sum_{i=1}^{m} \mathbb{E}_\sigma\left[\log_2 |N_{\sigma,\tau}(i)|\right]\right]$$

$$\leq \mathbb{E}_\sigma\left[\sum_{i=1}^{m} \mathbb{E}_\tau\left[\log_2 |N_{\sigma,\tau}(i)|\right]\right]$$

discrete

$$\leq \mathbb{E}_\sigma\left[\sum_{i=1}^{m} \frac{\log_2(1) + \cdots + \log(d_i)}{d_i}\right]$$

$$\leq \sum_{i=1}^{m} \frac{\log_2(d_i!)}{d_i} \qquad \blacksquare$$

2) Application: counting Hamiltonian paths in tournaments

Szele 1943: the random uniform tournament on n vertices has in expectation $n!/2^{n-1}$ Hamiltonian paths.

<u>Proof</u>: Let $X_{u,n}$ be the number of Hamiltonian paths starting at u. Let's show by induction on n that $\mathbb{E}[X_{u,n}] = \frac{(n-1)!}{2^{n-1}}$.

- this is true for $n=1$: $\mathbb{E}[X_{u,n}] = 1 = \frac{0!}{2^0}$
- $$\mathbb{E}[X_{u,n}] = \sum_{v \neq u} \mathbb{P}(u \to v) \cdot \mathbb{E}[X_{v,n-1} \mid u \to v]$$

$$\overset{\text{by induction}}{=} \sum_{v \neq u} \frac{1}{2} \cdot \frac{(n-2)!}{2^{n-2}} = \frac{(n-1)!}{2^{n-1}}$$

the number of Hamiltonian paths of a random transitive tournament is $\sum_u X_{u,n} = \frac{n!}{2^{n-1}}$. ∎

In particular there exists a tournament with this many Hamiltonian paths! How tight is it?

Szele 1943: $O(n!/2^{3n/4})$

Alon 1990: $O(n^{3/2}\, n!/2^n)$

Friedgut & Kahn 2005: $O(n^{3/2-\varepsilon}\, n!/2^n)$ for some small $\varepsilon > 0$

+ Improvements on the constant factors for the lower bound.

Alon 1990: Every n-vertex tournament has at most $O(n^{3/2} n!/2^{n-1})$ Hamiltonian paths.

<u>Proof</u>: We show that every n-vertex tournament has at most $O(\sqrt{n} \cdot n!/2^n)$ Hamiltonian <u>cycles</u>.

Alon's result then follows by observing that:

Lemma: Given an n-vertex tournament with P Hamiltonian paths, there exists an $(n+1)$-vertex tournament with at least $P/4$ Hamiltonian cycles.

<u>Proof of the Lemma</u>: Let T be an n-vertex tournament with P Hamiltonian paths.

Add a vertex v with a random adjacency. Let T' be the $(n+1)$-vertex tournament we obtain this way and X be the number of Hamiltonian cycles of T'.

$$\mathbb{E}[X] = \sum_{u,w \neq v} \text{Prob}(u \to v \to w) \cdot \#(\text{Hamiltonian paths of } T \text{ between } u \text{ and } w)$$

$$= \sum_{u,w \neq v} \frac{1}{4} \cdot \#(\text{Hamiltonian paths of } T \text{ between } u \text{ and } w)$$

$$= \frac{P}{4}. \qquad \square$$

- Let $A = (a_{ij})_{1 \leq i,j \leq n}$ be the $n \times n$ matrix s.t.

$$a_{ij} = \begin{cases} 1 & \text{if } u_i \to u_j. \\ 0 & \text{otherwise} \end{cases}$$

The permanent of A counts the number of partitions into cycles (1-factor).

Hence $\#$ hamiltonian cycles $\leq \operatorname{perm}(A) \underset{\text{Brégman}}{\leq} \prod_{i=1}^{n} (d_i!)^{1/d_i}$ $(*)$

where d_i is $\deg^+(u_i)$

- $g : n \in \mathbb{N} \mapsto (n!)^{1/n}$ is log-concave

i.e. $g(n)\, g(n+2) \leq g(n+1)^2 \quad \forall n \geq 0$

$$\left(\frac{\log g(n) + \log(g(n+2))}{2} \leq \log g(n+1) \right)$$

$\rightsquigarrow$ as $\sum_{i=1}^{n} d_i = \binom{n}{2}$, $\prod_{i=1}^{n} (d_i!)^{1/d_i}$ is maximal when all d_i differ by at most 1.

$$\operatorname{perm}(A) \leq \prod_{i=1}^{n} \left(\left\lceil \frac{n-1}{2} \right\rceil ! \right)^{1/\left\lceil \frac{n-1}{2} \right\rceil} \underset{\text{Stirling}}{=} O\left(\sqrt{n}\, n! / 2^n \right)$$

$\blacksquare$